2. Conceptual Foundations of the Institutional Grammar 2.0

IG 2.0 is premised on a set of syntactic definitions, conceptualizations of institutional statements, and assumptions regarding institutional statements, as well as various forms of nesting (statement-level and component-level nesting). While these definitions, conceptualizations, and assumptions generalize across levels of IG encoding, how they are captured depends on at which level the encoder is working. In this section, we will thus lay out these foundational syntactic definitions, concepts, and assumptions. We start by o ering complete definitions of IG components more generally, and then move to defining key institutional statement concepts, including the various notions of nested institutional statements and associated assumptions. We further provide principles and operational guidelines for the identification of statements in as far as they relate to the fundamental concept *Action Situation*, namely context characterizations embedded in institutional statements and identification of institutional statement types in the first place. Concluding this section, we organize the coding levels based on involved features, along with providing principal guidelines for the coding process. Note that most concepts in this section are introduced with a pragmatic focus on providing essential foundational understanding to perform the coding of institutional statements. For an extended treatment of the conceptual foundations, refer to Frantz and Siddiki (2022).

2.1. Syntactic De nitions of Institutional Statement Components

The IG structure as referred to in this document relies on the elementary syntactic components of regulative and constitutive statements highlighted in Tables 2 and 3. Definitions of these components that hold across IG 2.0 encoding levels are provided alongside each syntactic component.

Syntactic Element	De nition
Attributes	An actor (individual or corporate) that carries out, or is expected to/to not carry out, the action (i.e., Aim) of the statement. The Attributes component may also contain descriptors of the actor. The presence of this component is *necessary* for any regulative statement.
Deontic	A prescriptive or permissive operator that defines to what extent the action of an institutional statement is compelled, restrained, or discretionary. In institutional statements varying levels of prescriptiveness or strength thereof are represented with di erent terms or language that situate along a continuum. The presence of this component is *optional* for regulative statements.
Aim	The goal or action of the statement assigned to the statement Attribute. The presence of this component is *necessary* for any regulative statement.
Object	The inanimate or animate part of an institutional statement that is the receiver of the action captured in the Aim. Objects can be of *direct* or *indirect* nature. Indirect objects are objects that are a ected or targeted by the application of the Aim to direct objects. Objects can both be real-world entities, or abstract ones (e.g., beliefs, concepts, facts, procedures). Abstract objects here further include complex constructs, such as conditions/processes that the responsible actor is or will be acting upon. Such complex objects can potentially take the structural form of institutional statements themselves. The presence of this component is *optional* for regulative statements.
Context	The context component instantiates settings in which the focal action of a statement applies, or qualifies the action indicated in an institutional statement. The former type of Context is referred to as an *"Activation Condition"*. The latter type of Context is referred to as an *"Execution Constraint"*. Both can occur in a given institutional statement, including multiples of either type. The presence of this component is *necessary* for any regulative statement, but can be implied (as with the original institutional grammar). Where no explicit Activation Condition is specified, the context clause is by default "under all conditions". Where no explicit Execution Constraints are specified, the context clause is by default "no constraints". It is important to note that *Context* in institutional statements reflects the context specific to the coded statement (Statement Context), as opposed to capturing context in the wider sense, making reference to the context of the policy or the domain more generally.
Or else	A sanctioning provision associated with the action indicated in a particular institutional statement represented in a nested institutional statement (as defined in the following discussion). Where combinations of regulative and constitutive statements are applied (hybrid institutional statements), the consequence can be existential in kind (see Table 3 and Section 4.4). The presence of this component is *optional* for regulative statements.

Table 2: Definitions of Syntactic Elements for Regulative Statements

Syntactic Element	De nition
Constituted Entity	The entity being constituted, reconstituted, modified or otherwise directly a ected within an institutional statement as characterized by the constitutive function. This can be an entity as relevant in the institutional setting, or the policy itself. The presence of this component is *necessary* for any constitutive statement.
Modal	An operator that signals necessity or (im-)possibility of constitution or modification of system features captured in the constitutive function. In institutional statements varying levels of necessity or strength thereof are represented with di erent terms or language that situate along a continuum. The presence of this component is *optional* for constitutive statements.
Constitutive Function	An expression that relates the constituted entity and the institutional setting, e.g., by defining, establishing, or modifying the former, including the conferral of status. Where constituting properties exist, constitutive functions functionally link Constituted Entity and Constituting Properties. The presence of this component is *necessary* for any constitutive statement.
Constituting Properties	Constituting Properties parameterize the Constituted Entity via the Constitutive Function, and can both be physical or abstract in kind. Constituting Properties may or may not be present in constitutive statements. The presence of this component is *optional* for constitutive statements.
Context	The Context instantiates settings in which the focal Constitutive Function of a statement applies, or qualifies the function indicated in an institutional statement. The former type of Context is referred to as an *"Activation Condition."* The latter type of Context is referred to as an *"Execution Constraint."* Both can occur in a given institutional statement, including multiples of either type. The presence of this component is *necessary* for any constitutive statement, but can be implied (as with the original institutional grammar). Where no explicit Activation Condition is specified, the context clause is by default "under all conditions". Where no explicit Execution Constraints are specified, the context clause is by default "no constraints". It is important to note that *Context* in institutional statements reflects the context specific to the coded statement (Statement Context), as opposed to capturing context in the wider sense, making reference to the context of the policy or the domain more generally.
Or else	A consequence characterization associated with the non-fulfilment or -enactment of the Constitutive Function indicated in a separate institutional statement nesting on the leading institutional statement (as defined in the following discussion). Consequences, as understood in the context of constitutive statements, can be existential in kind (e.g., invalidating policy), as opposed to the non-existential sanctioning provisions on the regulative side. The presence of this component is *optional* for constitutive statements.

Table 3: Definitions of Syntactic Elements for Constitutive Statements

2.2. Institutional Statements

Capturing both the regulative and constitutive nature of statements, in IG 2.0 *an institutional statement describes expected actions for actors within the presence or absence of particular constraints, or parameterizes features of an institutional system.*

To accommodate a more comprehensive representation of structure and content, the construction of institutional statements in IG 2.0 requires a di erentiated understanding of institutional statements, both in terms of *atomic* and *nested institutional statements* of various forms.

2.2.1. Atomic Institutional Statements

Before highlighting notions of nested institutional statements, we reiterate the notion of an atomic institutional statement as the elementary form of an institutional statement: an *atomic institutional statement* is a statement of regulative or constitutive kind that contains the corresponding necessary components, alongside potential optional components other than the *Or else*, and in which none of the components contains multiple values (*component-level combinations*) or is otherwise nested (*component-level nesting* – see Section 2.2.3). Examples for component-level combinations are multiple attributes or combinations of aims in a given institutional statement. Examples for component-level nesting include context characterizations (with details to be introduced at a later stage) expressed as institutional statements themselves.

Illustrating this concept, the statement *"Organic farmers must commit to organic farming standards"* is *atomic* in kind. It identifies the following elements:

- Attributes: Organic farmers
- Deontic: must
- Aim: commit to
- Direct object: organic farming standards
- (Implied) Context: under all conditions.

It does not include an *Or else*, and neither of the components contains value combinations.[7]

In contrast, the statement *"Organic farmers must commit to their organic farming standards and accommodate regular reviews of their practices"* combines multiple distinctive activities and associated objects, namely *"commit to organic farming standards"* and *"accommodate regular reviews of their practices"*, as illustrated below by aggregating all phrases by component type (multiple phrases are separated by semicolon).

- Attributes: Organic farmers
- Deontic: must
- Aim: commit to; accommodate
- Direct Object:[8] organic farming standards; regular reviews of their practices
- (Implied) Context: under all conditions.

We refer to any such combination of component values as a *component-level combination*.

In consequence, this statement is *not atomic* in kind.

[7] Note that a component can contain phrases; however, the terms need to represent a semantic unit in the context of the concerned institutional statement.

[8] The distinction between direct and indirect object is discussed in greater detail at a later stage.

2.2.2. Nested Institutional Statements

Motivating the conception of nested institutional statements is the pragmatic observation that statements can reflect a linkage or embedding of multiple actors or regulated activities, reflecting complex configurational settings; rules in form are generally not expressed in the atomic form specified above. Lack of specificity of these linkages undermines the coder's ability to comprehensively, and accurately, capture institutional content for versatile downstream analysis.

IG 2.0 accommodates two forms of institutional statement nesting: *horizontal nesting* and *vertical nesting*. Generally, horizontal nesting allows for the representation of multiple institutional statements that convey co-occurring or alternative actions, actor or object involvement, i.e., any case in which multiple of the same syntax element occur in an institutional statement.

Generally, vertical nesting allows for the representation of multiple institutional statements that convey coupled actions that follow from one another in the form of a consequential relationship. It is particularly suited to representing the case of consequentially linked statements in which statement A delineates permitted, required, or forbidden activity (for regulative statements), or is constitutive in kind, and statement B delineates sanctions for non-conformance with statement A. In the IG 2.0 parlance, statement A is considered a "*monitored statement*," and statement B a "*consequential statement*." Horizontal nesting and vertical nesting are described in more detail below.

Horizontal Nesting: Horizontal nesting describes a logical combination of two or more statements to capture institutional content comprehensively. Exemplified in narrative form, a horizontally nested statement can combine two or more statements. Borrowing the example introduced earlier (*"Organic farmers must commit to their organic farming standards and accommodate regular reviews of their practices"*), horizontal nesting reflects the decomposition of such statement into two logically-combined atomic institutional statements, such as

"Organic farmers must commit to organic farming standards" **[AND]** *"Organic farmers must accommodate regular reviews of their practices"*

Horizontal nesting allows for more complex constructs linking an, in principle, arbitrary number of atomic institutional statements and combinations thereof, such as

("Organic farmers must commit to organic farming standards" **[AND]**
"Organic farmers must accommodate regular reviews of their practices") **[XOR]**
("Organic farmers must **[NOT]** *sell their produce under the organic farming label").*

In this example, we can observe the linkage of three statements, in which two AND-combined statements apply as an alternative (XOR) to a third statement.

Note the use of parentheses to signal the precedence of individual statements where multiple logical operators are used.

Possible logical operators for horizontal nesting, or *statement combinations*, are **[AND]** (conjunction), **[OR]** (inclusive disjunction; colloquially: **AND/OR**), or **[XOR]** (exclusive disjunction; colloquially: **EITHER/OR**). Where negation is involved, those can be combined with the operator **[NOT]** (as highlighted in the previous example). An alternative equivalent representation is

("Organic farmers must commit to organic farming standards" **[AND]**
"Organic farmers must accommodate regular reviews of their practices") **[XOR]**
[NOT] *("Organic farmers must sell their produce under the organic farming label").*

A common use case in practice is the decomposition of action combinations (e.g., *"... establish and maintain ..."*) or multiple actors (e.g., *"... producers or inspectors of the facility ..."*).[9]

[9] The decomposition of such component-level combinations is discussed as part of the coding guidelines in Section 4, speci cally in Table 7, Item *Decomposition of component level combinations*, as well as the same item in Table 8 for regulative statements. Table 10, Item *Decomposition of component level combinations* provides guidelines for

Vertical Nesting: Vertical nesting describes a relationship of two or more statements, in which the leading statement (*monitored statement*) describes an action that is regulated by a second statement nested in the Or else component (*consequential statement*), rendering the Or else an abstract component backed by a separate institutional statement. More accurately, the Or else does not reflect an IG component, but rather a logical connective that links two institutional statements. In this linkage, the second statement reflects a consequence of violating the instructions captured in the monitored statement. Consequences generally involve some pay-o for non-compliance or compliance.[1] Exemplifying vertical nesting in narrative form, we can write

"Organic farmers must comply with organic farming regulations", **OR ELSE** *"Certifiers must revoke the organic farming certification".*

Note that both forms of nesting can be combined, i.e., monitored and consequential statements can embed horizontal nesting. Extending the previous example, we can state (visually supported by indentation to signal the corresponding forms of nesting)

("Organic farmers must comply with organic farming regulations" **AND**
"Organic farmers must accommodate regular review of their practices"),
OR ELSE *("Certifiers must suspend the organic farming certification"* **XOR**
"Certifiers must revoke the organic farming certification").

Note the use of parentheses to signal precedence of the respective statements. Vertical nesting can occur across an arbitrary number of levels (i.e., a consequential statement may be a monitored statement in deeper levels of nesting). Exemplifying multi-level nesting, we can state

("Organic farmers must comply with organic farming regulations" **AND**
"Organic farmers must accommodate regular review of their practices"),
OR ELSE *("Certifiers must suspend the organic farming certification"* **XOR**
"Certifiers must revoke the organic farming certification"),
OR ELSE *"USDA may revoke certifier's accreditation".*

The combination of both forms of nesting a ords the representation of complex institutional arrangements, both in terms of institutional content (horizontal nesting) and enforcement (vertical nesting). The principles are schematically highlighted in Figure 1.[11] While exemplified here for regulative statements, the principles equally apply to constitutive, and in consequence, hybrid institutional statements (see Section 4.4).

2.2.3. Component-Level Nesting

In addition to the nesting of statements, IG 2.0 further recognizes the possibility of nesting on individual components. Specific components may be substituted with entire institutional statements, or generalizable instances of activity (e.g., statements of fact, observed behavior) expressed in the syntactic form of institutional statements. For regulative statements, those may characterize individuals with respect to their actor properties, embed statements as abstract objects (e.g., the belief that citizens must comply with the law), or express preconditions, i.e., statements whose fulfillment is a precondition for the application of a given statement (as exemplified below).

To motivate this approach explicitly, we will use the following example: *"Organic farmers may sell their produce under the organic label under the condition that organic farmers apply for certification}"*, with the precondition (the *Activation Condition* as a more specific variant of the Context component,

constitutive statements. General operational guidelines for ne-grained decomposition can be found in Table 9, Item *Logical relationships among statement components*. Note: All preceding table item references are clickable links.

[10] Where constitutive statements are involved, consequences can further be existential in kind.

[11] An formal discussion of nesting can be found in Frantz et al. (2013) and an extended conceptual discussion is provided in (Frantz & Siddiki, 2022).

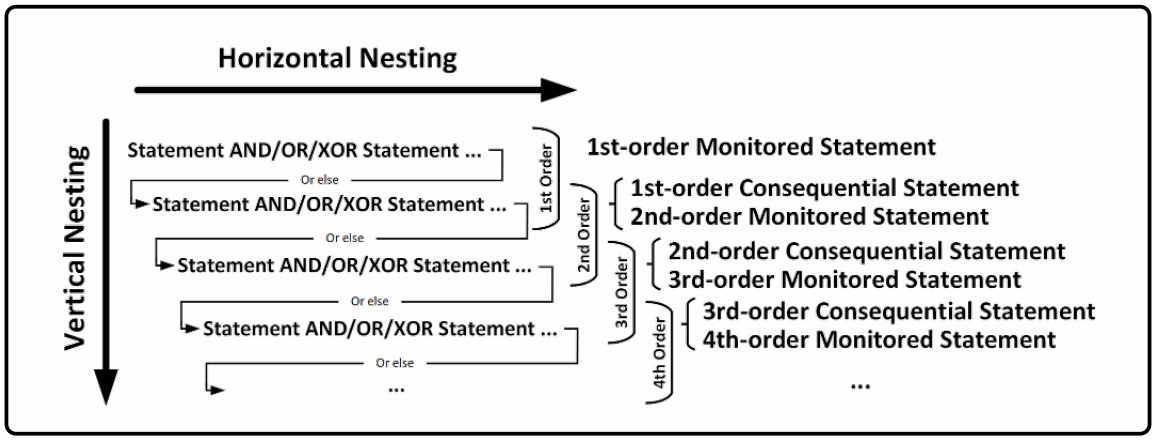


Figure 1: Institutional Grammar Nesting Principles

as to be discussed in Section 2.4) identified in braces. Rewriting this statement for illustration, we find the following component characterization:

- Attributes: Organic farmers
- Deontic: may
- Aim: sell
- Direct Object: their produce
- Execution Constraint (Context): under the organic farming label
- Activation Condition (Context):[12] under the condition that organic farmers apply for certification

Here the instance of behavior embedded in the activation condition reflects the precondition, but, upon closer inspection, may in itself be expressed in terms of syntactic components of institutional statements, either reflecting a specific action configuration (as statement of fact, behavioral instance, or observation) or an institutional statement in its own right (including combinations of statements).

We refer to cases where individual components of an institutional statement can be replaced by institutional statements (or statements that follow the syntactic structure of institutional statements) as *component-level nesting*. An institutional statement that contains any such nested structure (as initially indicated in Section 2.2.1) is not atomic in kind.

Expanding the statement above correspondingly, we arrive at the following structure exemplifying the nesting on the activation condition:

- Attributes: Organic farmers
- Deontic: may
- Aim: sell
- Direct Object: their produce
- Execution Constraint (Context): under the organic farming label
- Activation Condition (Context): under the condition that
  - Attributes: organic farmers
  - Aim: apply

[12] The activation condition and execution constraint components are a specialization of the Context component, which will be introduced at greater detail in Section 2.4.

  - Direct Object: for certification

The use case for such component-level nesting further includes the articulation of abstract concepts, including non-physical concepts (e.g., beliefs, suspicions, etc.) about individuals' behaviors inasfar as those are relevant to capture the institutional setting accurately (e.g., an official may sanction an individual if the official believes that the individual has performed a violation).

Exemplifying nesting on the object component, we can use the statement *"Inspectors must ensure that organic farmers comply with organic farming regulations"*. In this instance, the nesting occurs on the object component, i.e., the receiver of the action *ensure*, as visualized below.

- Attributes: Inspector
- Deontic: must
- Aim: ensure
- Direct Object:
  - Attributes: organic farmers
  - Aim: comply
  - Direct Object: organic farming regulations
- (Implied) Context: under all conditions

While not motivated further at this stage, component-level nesting (as statement-level nesting) equally applies to regulative and constitutive statements. Component-level nesting can apply to Attributes, Objects, Context components (Activation conditions and Execution constraints) in regulative statements, as well as to Constituted Entity, Constituting Properties and Context components in constitutive statements. It can further be applied in combination with all forms of statement-level nesting, including the further nesting within component-level nested statements. (For example, imagine a chain of preconditions expressed in nested statements).

The coding for all forms of nesting and combinations thereof, as conceptually described here, is specified in Section 4.

2.3. Object-Property Hierarchy

In addition to the nesting concepts, advanced coding relies on decomposing actors and objects into core descriptors and associated properties. For this purpose, we rely on the conceptual representation of an *Object-Property Hierarchy* as exemplified in Figure 2. In this visualization, statements such as *"... a written notification of proposed suspension or revocation of certification ..."* reflect an involved Object hierarchy centering on the *"notification"*, that has a property *"written"*. Looking at the context of the notification we recognize the concept of *"certification"* that has the property of being *"suspended"* or *"revoked"*, expressed as dependent Objects (*"suspension"*, *"revocation"*), whereas the latter concepts themselves have a shared property of being *"proposed"* in the first place. However, while the property *"written"* functionally depends on the *"notification"*, that is, writtenness alone does not make sense with an Object it refers to, the existence of a certification does not rely on the notification (i.e., it is functionally independent), and has a self-contained property hierarchy (suspended, revoked, proposed) as described above.

Interpreting complex Object specifications with this decomposition hierarchy in mind a ords a uniform coding approach. The structure of this hierarchy for the specific statement is shown in Figure 2 (the dashed line signals relationships between functionally independent Objects). Note specifically the

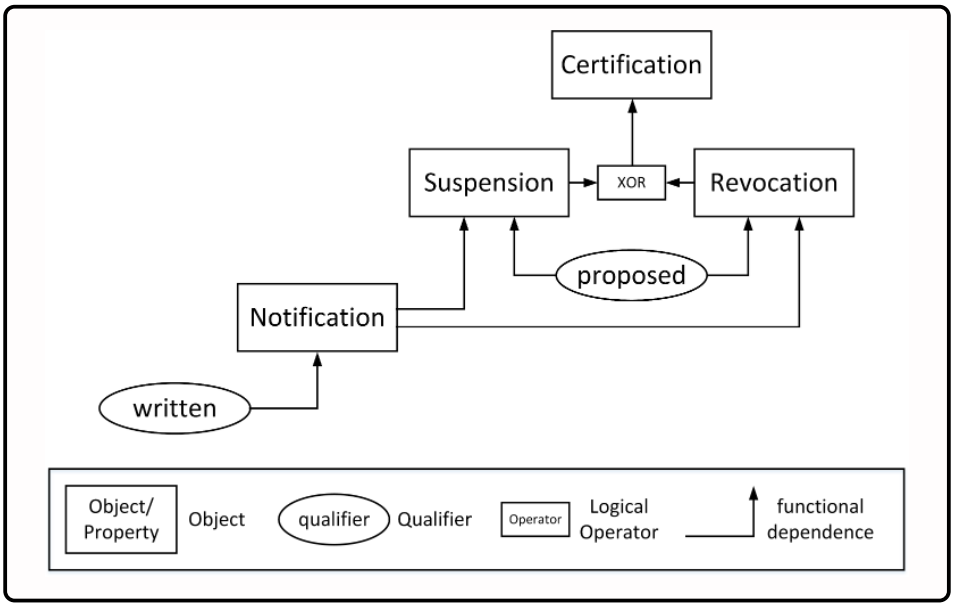


Figure 2: Object-Property Hierarchy for the given Example

potential use of logical operators (*"XOR"*), as well as the ability to reflect shared properties (*"proposed"*), to disambiguate the logical relationship between the identified properties, an aspect that is implicit in the textual representation.

Capturing all potential decomposition approaches, we can apply this scheme to functionally dependent properties (*"written"* in the previous example), functionally independent Objects or properties (*"certification"* in the previous example), and furthermore a ord the substitution of Objects by complete institutional statements. Since the latter aspect relies on richer contextualization, it will be exemplified in the context of the coding instructions. The stylized general form of the *Object-Property Hierarchy* is shown in Figure 3. Note that logical operators apply to both functionally dependent and independent properties and on any level of decomposition.

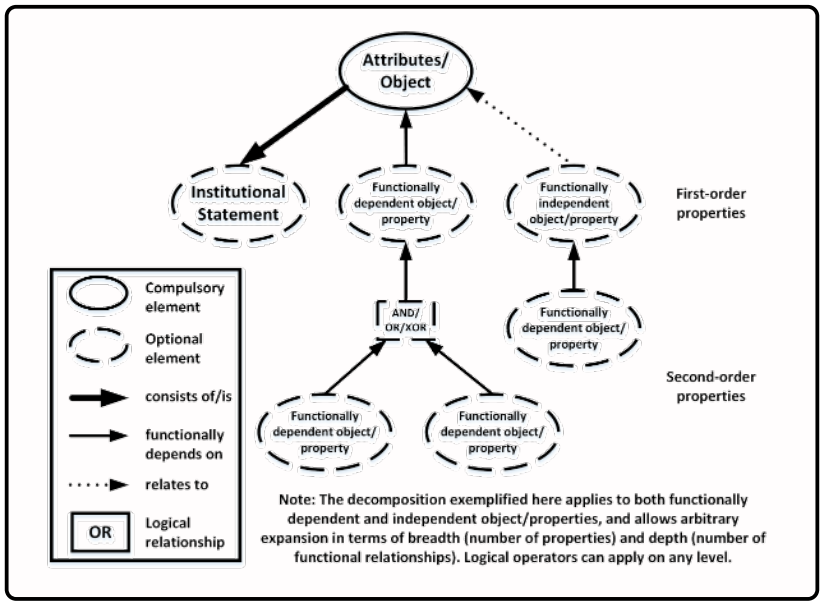


Figure 3: Object-Property Hierarchy